Victor Geadah

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Positions

2026– **Flatiron Institute – Center for Computational Neuroscience**, New York City, NY
Flatiron Research Fellow.

Education

2021–2026 **Princeton University**, Princeton, NJ
Ph.D. in Applied and Computational Mathematics
- Supervised by Jonathan W. Pillow (Princeton Neuroscience Institute).

2020–2021 **University of Cambridge**, Cambridge, UK
MASt in Applied Mathematics (Part III)

2017–2020 **Université de Montréal**, Montréal, Canada
B.Sc. in Pure and Applied Mathematics

Research Experience

2021–2026 **Princeton Neuroscience Institute**, Princeton, NJ
Graduate Researcher. Advisor: Jonathan W. Pillow

Summer '25 **Meta – CTRL-Labs**, New York City, NY
Research Scientist Intern.

Summer '24 **Flatiron Institute – Center for Computational Neuroscience**, New York City, NY
Summer Pre-Doctoral Researcher. Supervisor: Alex H. Williams

2019–2021 **Mila - Quebec Artificial Intelligence Institute**, Montréal, Canada
Undergraduate Researcher. Advisors: Guillaume Lajoie, Guy Wolf

Summer '18 **Université de Montréal**, Montréal, Canada
Undergraduate Researcher. Advisor: Iosif Polterovich

Awards & Scholarships

2025–2026 Porter Ogden Jacobus Fellowship – Princeton University
→ Princeton University's highest honor for graduate students in their final year.

2021–2025 Doctoral (B2X) Research Scholarship – FRQNT

2021–2024 Canada Graduate Scholarship - Doctoral (CGS-D) – NSERC

2020 Jean-Maranda Prize – Université de Montréal
→ Graduating student with highest academic ranking in pure and applied mathematics

2020 Undergraduate Introduction to Research Scholarship – IVADO

2020 Honorable Mention – Mathematical Contest in Modeling (MCM)

2019 Undergraduate Summer Scholarship – CRM-ISM

2018 Undergraduate Summer Research Award (USRA), with Supplement – NSERC, FRQNT

2018 Royal Bank of Canada Scholarship Program – RBC

Publications & abstracts

Journal Articles

1. **Geadah**[*], V., Barello[*], G., Greenidge, D., Charles, A. S., and Pillow, J. W. "Sparse-Coding Variational Autoencoders". In: *Neural Computation* 36.12 (2024), pp. 2571–2601. [link | code].
2. **Geadah, V.**, Horoi, S., Kerg, G., Wolf, G., and Lajoie, G. "Neural networks with optimized single-neuron adaptation uncover biologically plausible regularization". In: *PLOS Computational Biology* 20.12 (2024), e1012567. [link | code].

[*]: These authors contributed equally.

Conference Proceedings

3. **Geadah, V.**, Nejatbakhsh, A., Lipshutz, D., Pillow, J. W., and Williams, A. H. "Modeling neural activity with Conditionally Linear Dynamical Systems". In: *NeurIPS*. 2025. [link | code].
4. Liu, Y. H., **Geadah, V.**, and Pillow, J. W. "Flexible inference of learning rules from de novo learning data using neural networks". In: *NeurIPS*. 2025. [link].
5. Nejatbakhsh, A., **Geadah, V.**, Williams, A. H., and Lipshutz, D. "Comparing noisy neural population dynamics using optimal transport distances". In: *ICLR*. 2025. [**oral** | link].
6. **Geadah, V.**, Arbelaiz, J., Ritz, H., Daw, N. D., Cohen, J. D., and Pillow, J. W. "Inferring system and optimal control parameters of closed-loop systems from partial observations". In: *63rd IEEE Conference on Decision and Control (CDC)*. 2024, pp. 8006–8013. [talk | link | pdf].
7. **Geadah, V.**, Laboratory, I. B., and Pillow, J. W. "Parsing neural dynamics with infinite recurrent switching linear dynamical systems". In: *ICLR*. 2024. [link].
8. Horoi, S., **Geadah, V.**, Wolf, G., and Lajoie, G. "Low-Dimensional Dynamics of Encoding and Learning in Recurrent Neural Networks". In: *Advances in Artificial Intelligence*. Springer International Publishing, 2020. [link].

Preprints & Work Under Review

9. **Geadah, V.** and Pillow, J. W. "Inferring learning rules during de novo task learning". Under review. 2025.

10. Jha, A., **Geadah, V.**, and Pillow, J. W. "Modeling complex animal behavior with latent state inverse reinforcement learning". 2024.

Conference Abstracts

11. Liu, Y. H., **Geadah, V.**, and Pillow, J. W. "Flexible inference of animal learning rules using neural networks". In: *COSYNE*. 2026. [poster].
12. **Geadah, V.**, Nejatbakhsh, A., Lipshutz, D., Pillow, J. W., and Williams, A. H. "Modeling Neural Activity with Conditionally Linear Dynamical Systems". In: *SAND*. 2025. [**talk**].
13. **Geadah, V.** and Pillow, J. W. "Inferring single-animal learning objectives in mice decision-making". In: *COSYNE*. 2025. [poster].
14. **Geadah, V.**, Nejatbakhsh, A., Lipshutz, D., Pillow, J. W., and Williams, A. H. "Capturing condition dependence in neural dynamics with Gaussian process linear dynamical systems". In: *COSYNE*. 2025. [poster].
15. Nejatbakhsh, A., **Geadah, V.**, Williams, A. H., and Lipshutz, D. "Comparing noisy neural population dynamics using optimal transport distances". In: *COSYNE*. 2025. [poster].
16. **Geadah, V.** and Pillow, J. W. "Parsing neural dynamics with infinite recurrent switching linear dynamical systems". In: *COSYNE*. 2023. [poster].
17. **Geadah, V.** and Pillow, J. W. "Non-exchangeability in Infinite Switching Linear Dynamical Systems". In: *NeurIPS Workshop on Gaussian Processes, Spatiotemporal Modeling, and Decision-making Systems*. 2022. [poster | pdf].
18. **Geadah, V.**, Horoi, S., Kerg, G., Wolf, G., and Lajoie, G. "Top-down optimization recovers biological coding principles of single-neuron adaptation in RNNs". In: *COSYNE*. 2022. [poster].
19. **Geadah, V.**, Horoi, S., Kerg, G., Wolf, G., and Lajoie, G. "Network-level computational advantages of single-neuron adaptation". In: *COSYNE*. 2021. [poster].
20. **Geadah, V.**, Wolf, G., and Lajoie, G. "Single neuron nonlinearities and their impact on learning dynamics of recurrent networks". In: *NAISys*. 2020. [**talk**].
21. **Geadah, V.** and Polterovich, I. "Asymptotics of Steklov Eigenvalues for Regular Polygons". In: *Annual Symposium in Mathematics for a Future in Research and Industry*. 2019. [poster].

    → **Poster Public Prize** by the *Faculty of Graduate Studies and Post-Doctoral Research*.

Teaching

Spring '24 — **Princeton University**, Princeton, NJ
Teaching Assistant, Statistical Modeling and Analysis of Neural Data (NEU 560)
- Guest lecture on Monte Carlo Integration and Importance Sampling

Fall '19 — **Université de Montréal**, Montréal, Canada
Teaching Assistant, Real Analysis I (MAT1000)

Academic Service

Princeton University, Princeton, NJ

2023– *Mentor*, COSYNE Undergraduate Travel Grant Program

2022–2025 *Mentor*, PACM Undergraduate Certificate

Université de Montréal, Montréal, Canada

2018–2020 *Co-founder*, Mathematical Modeling Club

2018–2020 *Organizer*, Club Mathématique

2018–2020 *Mentor*, Sensibilisation aux Études, à l'Université et à la Recherche (SEUR)

Reviewer: ICLR 2025, NeurIPS 2025, *Neural Computation*

Invited Talks

2026 Learning Club, Computational Machinery of Cognition – Safavi Lab, TU Dresden (online)

2025 COSYNE "Dynamics of brain computations through the lens of control theory" workshop

Princeton Neuroscience Institute Seminar, Princeton University

2024 PACM Graduate Student Seminar, Princeton University

NeuroStatsLab Meeting, Flatiron Institute – Center for Computational Neuroscience

Cohen Lab Meeting, Princeton University

2023 SAMARI Symposium, Université de Montréal

2022 P6 Seminar (invited by Jonathan Cohen), Princeton University

SAMARI Symposium, Université de Montréal

2020 Séminaire d'été du Département de Mathématique et Statistique, Université de Montréal

2019 Waterloo Mathematics Undergraduate Research Conference, University of Waterloo

Séminaire d'été du Département de Mathématique et Statistique, Université de Montréal

Canadian Undergraduate Mathematics Conference (CUMC), Queen's University

2018 Séminaire d'été du Département de Mathématique et Statistique, Université de Montréal

Dates in *italic* are forthcoming.
Last compiled: October 2026